ModelIt!

Teacher's Guide

How Do Animals Use Their Body Parts?

The Science of Structure and Function

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-LS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-LS1-1

Use materials to design a solution to a human problem by mimicking how plants and animals use their external parts to help them survive, grow, and meet their needs.

Three-Dimensional Learning

Science Practice	**Constructing Explanations** — Students explain how specific body parts help animals survive by matching structure to function.
Disciplinary Core Idea	**LS1.A Structure and Function** — All organisms have external parts that help them survive, grow, and meet their needs.
Crosscutting Concept	**Structure and Function** — Students explore how the shape and size of body parts relate to the jobs they perform.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Animals have amazing body parts that are perfectly designed for the jobs they need to do. From eagle talons to elephant trunks, every structure has a function.

Structure Matches Function

In science, structure means the shape or form of something, and function means the job it does. A bird's wings are shaped for flying — they are wide, flat, and lightweight. A fish's fins are shaped for swimming — they push water to help the fish move. A cat's claws are curved and sharp for climbing and catching. Every animal body part has a shape that matches its job perfectly.

Different Body Parts for Different Jobs

Animals use their body parts to meet their basic needs: getting food, staying safe, and moving around. A frog's long sticky tongue catches flies. A turtle's hard shell protects it from predators. A cheetah's long legs help it run fast to catch prey. Even the same body part can look different in different animals — an eagle's beak is hooked for tearing meat, while a hummingbird's beak is long and thin for sipping nectar. The shape tells us what the animal eats and how it lives.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify body part size as something that varies between animals (external) and ability to get food as something that depends on the body part (internal).

E — **Establish Relationships**
Students discover that when a body part better matches the food source, the ability to get food goes up.

V — **Visualize & Model**
Students build a simple model showing how body part size connects to the ability to get food.

E — **Evaluate Outcomes**
Students test their model by comparing long-beaked and short-beaked scenarios with deep flowers.

R — **Revise & Extend**
Students reflect on how their model explains why different birds have different beak shapes in nature.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at this hummingbird's beak! It is so long and thin. Why do you think it looks like that? Let us find out!"*

Watch For

Show photos of three different animal body parts. Ask: What job does each one do?

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn two important words: STRUCTURE means a body part's shape. FUNCTION means the job it does."*

Watch For

Practice: Point to your hand (structure) and show grabbing (function). Point to legs (structure) and show walking (function).

Slide 3: Let's Be Scientists!

3 min

Say: *"How do animals use their body parts to survive? Why does a bird's beak look different from a dog's mouth?"*

Watch For

Show three different bird beaks. Students predict what each bird eats based on beak shape.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Body Part Size is how big or long the beak is. Ability to Get Food is how well the bird can eat."*

Watch For

Show a long straw and a short straw. Ask: Which one could reach the bottom of a tall glass?

Slide 5: How Are They Connected?

5 min

Say: *"When the beak is longer, can the bird reach deeper into a flower? Yes! Longer beak means better ability to get nectar."*

Watch For

Students try picking up small items with long tweezers vs. short tweezers to feel the difference.

Slide 6: What Did We Find Out? 5 min

Say: *"We found out that body parts have special shapes that match their jobs. A long beak is perfect for reaching deep flowers!"*

Watch For

Review discoveries. Match three animals to their body parts and functions as a class.

Slide 7: STEM Challenge 10 min

Say: *"Now YOU will design a tool inspired by an animal body part. Can you build something that picks up small beads?"*

Watch For

Distribute clothespins, spoons, pipe cleaners, cups of beads. Students design and test grabbing tools.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Body Part Size
- **INTERNAL:** Ability to Get Food

Relationships

- **Body Part Size to Ability to Get Food:** POSITIVE (+) — A longer beak helps the bird reach food in deep flowers better. When the body part better matches the food source, the animal is better at getting food.

Key Discoveries

- Animals have special body parts that help them survive
- The shape and size of a body part matches the job it needs to do
- A long beak helps birds reach food that short-beaked birds cannot reach

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Animals choose to grow body parts they need"	Animals do not choose their body parts. They are born with them! Over many, many years, animals that had body parts that helped them survive had more babies, and those babies inherited the same helpful body parts.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All beaks do the same job"	Different beak shapes do different jobs. A pelican's wide beak scoops fish from water. A woodpecker's strong pointy beak hammers into trees. A parrot's curved beak cracks open hard nuts. The shape tells us what the bird eats!